GRDD: A Dataset for Greek Dialectal NLP

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Abstract

This paper presents GRDD[1], a dataset for the computational study of Modern Greek dialects. It consists of raw text data from four dialects of Modern Greek: Cretan, Pontic, Northern, and Cypriot Greek. The dataset is of considerable size, albeit imbalanced, and presents an attempt to create large scale dialectal resources of this type for Modern Greek dialects. We use the dataset in a dialect identification task, experimenting with both traditional Machine Learning (ML) algorithms as well as simple Deep Learning (DL) architectures. The results show a very good performance on the identification task, suggesting that the dialects in question possess distinctive characteristics that enable even simple ML models to perform well. Error analysis conducted on the top-performing algorithms reveals that many errors can be attributed to inadequate dataset cleaning.

Keywords: Modern Greek dialects, dialectology, dataset creation, Machine Learning, Natural Language Processing, Neural Networks, Dialect Identification

1. Introduction

A number of computational resources have been developed for Standard Modern Greek (SMG) in the recent years (see Papantoniou and Tzitzikas (2020) for an overview). However,

[1] The dataset can be found here: https://github.com/StergiosCha/Greek_dialect_corpus.

the dialects of Modern Greek have not received comparable attention in Natural Language Processing (NLP). One major reason for this is the lack of large scale dialectal resources that can be used to train NLP models for Greek dialects.[2] Acquiring such dialectal data is often an expensive process. Nevertheless, several Modern Greek dialects have a presence online, and various data sources can be identified and extracted. For instance, Cypriot Greek (CG) data can be found in numerous online forums, social media platforms, and blog posts. While it is more challenging to find online data for Pontic and Cretan Greek, some sources still exist, including traditional story narrations, song lyrics, older literary texts in the dialect, and attempts to translate classical texts into these dialects. Similar considerations apply to the Northern varieties.

In this paper, we try to identify freely available online sources of dialectal data for four Modern Greek dialects. After these sources are located, we extract and lightly preprocess the data, resulting in a substantial but imbalanced dataset: 2.3 million words from Cypriot Greek (CG), 880,000 words from Cretan, 282,000 words from Pontic, and 35,000 words from Northern Greek. To ensure data quality, we randomly sample 10,000 words from each dialect and have native speakers rate their validity. Finally, we utilize the dataset for a dialect identification task and present our findings.

2. Related work

Datasets for training ML and DL algorithms and models on Greek dialectal data are sparse. There is a small dataset for the Griko (Anastasopoulos et al., 2018) dialect[3], which is parallel (Italian-Griko) and annotated with Part-of-Speech (POS) tags. Sababa and Stassopoulou (2018) design a simple Naive Bayes Algorithm to distinguish between Standard Modern Greek (SMG) and Cypriot Greek (CG) data and terst it on a dataset of around 7k CG sentences. Themistocleous (2019) uses a simple DNN to distinguish between SMG and CG based on acoustic properties only (in fact, via a single sonorant). The results present an improvement over classical ML algorithms. Hadjidas and Vollmer (2015) present the Cypriot version of the Multi-CAST corpus, a dataset of annotated spoken texts. Lastly, Karasimos et

[2] There is vast dialectal material from theoretical work in dialects, but these resources are not fit at their present form to be used for training ML models. Turning all these resources into computational resources has varying degrees of difficulty and it is not the purpose of this paper to discuss this issue.

[3] Griko is a Modern Greek dialect spoken in Southern Italy (Rohlfs and Sicuro, 1977; Katsoyannou, 1995; Chatzikyriakidis, 2010)

al., 2008 present an attempt to provide a database with linguistic and meta-linguistic corpora based on recorded dialectal material (505 hours of recordings).

3. The varieties

We collected data from 5 varieties of Modern Greek: Standard Modern Greek (SMG), Cypriot Greek, Pontic Greek, Cretan and Northern Greek. A few words about these varieties are in order.

3.1 Standard Modern Greek

Standard Modern Greek (SMG) is the official language of the Greek and Cypriot state. According to (Mackridge, 1985) has its basis in the Peloponnese Greek dialects. This can be traced back to the Greek war of Independence (1821-9) where one of the first areas to be liberated was the Peloponnese, as well as the fact that the Peloponnese dialects (with the exception of Tsakonian) were the dialects closer to the written language. This newly formed variety was further enriched by elements of the dialects of dominant Greek communities of the time, notably Greeks from Istanbul and the Ionian islands. Other factors like compulsory education, military service, and the the advent of radio and television "have made Greece into a melting pot in which speakers of various kinds of Greek have gradually sunk their linguistic differences" as Mackridge (1985) notes.

3.2 Cypriot Greek

Cypriot Greek is the Greek dialect spoken by Greek Cypriots, as well as some Turkish Cypriots. The first evidence of Cypriot Greek comes from a legal text dated in the 14th century (The Assizes)[4]. Despite being the native language of most (if not all) Cypriot Greeks in both Cyprus and the diaspora, the official language of the Cypriot Greek state is SMG (Newton, 1972; Tsiplakou, 2009, 2014). This means that modern day CG stands in a diglossic relationship to SMG (Tsiplakou, 2009, 2014). Tsiplakou (2009) further shows that this diglossic relationship in modern day Cyprus is between SMG and an emergent pancypriot Koine. Cypriot Greek has a number of distinct features compared to SMG. We indicatively mention the distinct weak object pronoun positioning system (Agouraki, 2001;

[4] Note that the presence of Greek in Cyprus dates back to the 11-10th century BC (Arcadocypriot). However, Cypriot Greek is not an evolution of the Arcadocypriot dialect of Ancient Greek but rather evolved from Koine Greek.

Chatzikyriakidis, 2012), as well as the use of clefts to express wide focus constructions (Grohmann et al., 2006; Grohmann, 2009).

3.3 Pontic Greek

We use the term Pontic Greek (PG from now on) to refer to the dialect as it is spoken today in Modern day Greece, since a form of Pontic, Romeyka Pontic is still spoken today in some villages of Trabzon and the area in modern day Turkey (Sitaridou and Chatzikyriakidis, 2012; Sitaridou, 2013; Schreiber, 2018). The roots of PG, similarly all other Modern Greek dialects (with the exception of Tsakonika), can be traced back to Koine Greek, i.e. the form of Greek used during the Hellenistic years. However, as Mackridge (1987) notes, PG came to be markedly distinct to the other Greek dialects probably due to the Seljuk invasion of the 11th century that basically split the Pontus area from the other areas of the Byzantine empire[5]. PG spoken in the Pontus region was not a uniform dialect but included several subdialects. A number of classifications have been proposed by the years (Papadopoulos, 1955; Triandafyllidis, 1981). However, all these sub-dialects tend to or have already been neutralized in favour of a PG koine (Tombaidis, 1996; Χατζησαββίδης, 1995). The population exchange brought 1.5 million Pontic Greeks into mainland Greece, mainly in the regions of Macedonia and Thrace. At that point the contact between PG and SMG begins. The fact that the PG dialect was treated with hostility from the official Greek state of the time as well as the need for maintaining a distinct cultural identity led to the creation of a PG koine neutralizing sub-dialectal idiosyncrasies. Furthermore, Pontic Greeks from different areas of Pontus ended up living together in mainland Greece. This last fact, as well as the tendency of PG people to prefer marriages with their own kind, played an important role in the development of a koine form. PG has some interesting divergences from the linguistic structure as found in SMG. Some of them include the topicalization particle pa (Sitaridou and Kaltsa, 2014), an across the board enclitic system for weak object pronouns (Chatzikyriakidis, 2010), and the allowance of multiple wh-fronting among other characteristics (Michelioudakis and Sitaridou, 2016).

[5] PG along with Cappadocian Greek comprise the Asia Minor Greek dialectal group.

3.4 Cretan Greek

Cretan Greek is a Greek dialect spoken in the island of Crete, also derived from Koine Greek (Mackridge, 1985). According to traditional classification, the dialect belongs to the south-western dialectal group of Greek dialects (Hatzidakis, 1905). Cretan Greek has some compelling features that differentiate it among the other Greek dialects. The most characteristic one is the substitution of palatoalveolars for palatalized velars (Mackridge, 1985). The so-called 'Cretan-type' fronting is a process in which the articulation is characterized as 'prepalatal', i.e. midway between a palatal and an alveolar realization (Manolessou and Pantelidis, 2012).

3.5 Kozani/Grevena Greek

This is the form of Greek as it is spoken in the areas of Kozani, Grevena in northwestern Greece. These dialects belong to a bigger group, i.e. that of the Northern dialects. These also include the dialects spoken in Thessaly and the Northeast Aegean, among other places. Some of the characteristics of the dialects include the use of prepositional genitives to denote possession (Chatzikyriakidis et al., 2023) and unstressed mid-vowel raising and unstressed high-vowel deletion, among other characteristics (Ntinas, 2005; Θαβωρης, 1994).

4. The dataset

We focused on data that are freely available on the web. These include dialectal data from blogs, websites, as well as freely available literary texts. In the case of SMG, we just used a subset of the Greek wikipedia corpus using the Greek wikipedia dump.[6] The first step in the collection process involved the identification and collection of urls by native speakers of the dialects or people with theoretical knowledge of the dialects, where such data can be found. We split ourselves into four groups, each focusing on finding tokens on the web. We then developed several Python scripts to extract the dialectal text in raw format, followed by applying a series of basic preprocessing steps to clean the dataset of Greek dialects (Cretan, Pontic, Cypriot, Northern, and Modern Greek). We removed blank lines, special characters,

[6] https://dumps.wikimedia.org/backup-index.html

duplicates, punctuation, and extra white spaces. The final version of the dataset used in this study (v1) was formatted into CSV files, with each column representing a different dialect and labeled accordingly. The number of words for each dialect category after the cleaning process is shown in Table 1.

Table 1. The table shows the number of words for each dialect in the dataset.

Dialect	Number of words
Standard Modern Greek	6,5m
Cretan	880k
Pontic	282k
Northern	35k
Cypriot	1,3m

To verify the validity of the collected data, we extracted a random sample of 10k lines for each dialect. The evaluation was conducted using a scale from 1 to 10, based on their dialectal intuitions (see Table 2).

Table 2. The table shows the average score of the evaluation of the data by native speakers.

Dialect	Avg Score
Cretan	8
Pontic	7.5
Northern	9.5
Cypriot	7

5. Dialect Identification

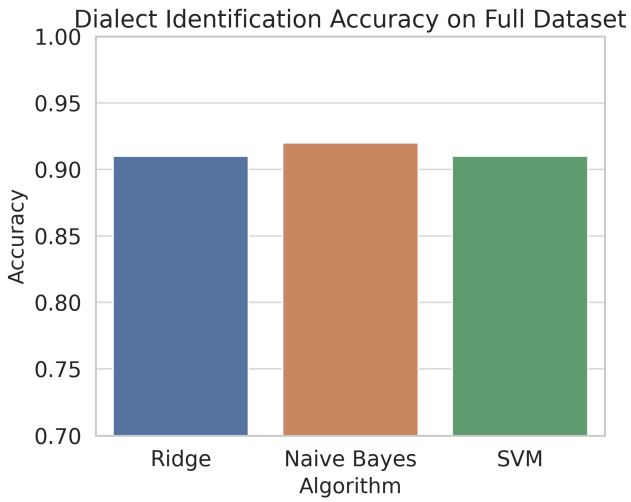
We utilized the dataset for Dialect Identification using: a) classic Machine Learning (ML) algorithms, and b) a vanilla BiLSTM model. We applied both approaches to the full

imbalanced dataset, as well as to three balanced subsets of the original dataset, each containing 5k, 20k, and 20k lines per dialect, respectively.

5.1 Classic ML algorithms

We ran a number of very classic ML algorithms using a simple word-gram language model (both 1-gram and 3- gram). We see that accuracy for the full dataset is quite high, which seems to suggest that the models are able to correctly identify distinctive features for the dialects and learn to classify them correctly. The results are reported below:

Figure 1. Top-three performing algorithms on the DI task (Ridge-blue, Naive Bayes- green, and SVC-red)



We created three versions of the original dataset with 5k, 20k and 80k lines respectively to investigate the effect of data imbalance in the classification. We present the results of the top three performing algorithms for each dataset below:

Table 3. Top three performing algorithms on three balanced datasets

Dataset/Model	Ridge	NB	SVC
5k	**0.76**	**0.76**	0.75
20k	**0.82**	**0.81**	0.82
80k	0.89	**0.90**	0.88

We see a steady increase in accuracy for all algorithms as more data are added. With 80k lines per each dialect we get an accuracy of 0.90 for the NB model. This is getting close to the performance on the original imbalanced dataset (NB performs 0.94 accuracy).

5.2 BiLSTM

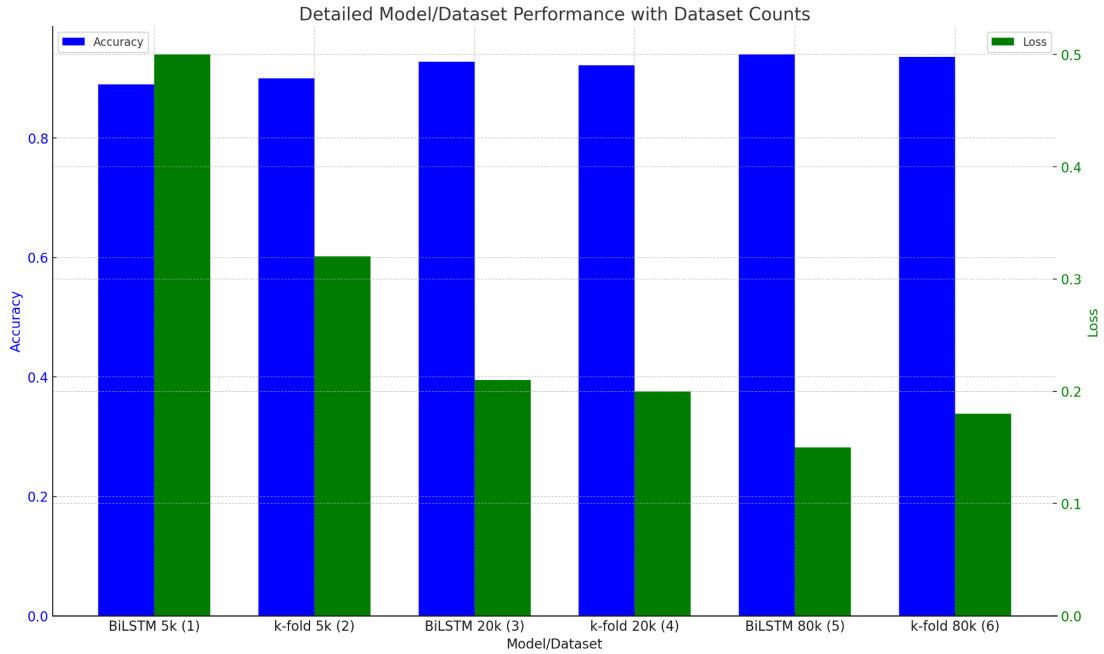
Following the relative success of simple ML algorithms in the task, we experimented with a more elaborate Deep Learning (DL) model, in particular a vanilla BiLSTM model, which consists of an 128 embedding layer, two BiLSTM layers (64 and 32 layers respectively), a densely connected network with 64 units and relu as the activation function and, lastly, a final layer with softmax activation. We first run this model using only two epochs. The results present a very high accuracy of 97% and low loss (approximately 10%). To check for model overfitting, we performed k-fold validation using the same number of epochs. The results show a decrease in average accuracy as well as an increase in average loss. However, the results seem to indicate that the model is generalizing well to the training data, even though some overfitting seems to be at play. The results are shown in:

Table 4. Comparison of BiLSTM and k-fold

	Accuracy	**Loss**
BiLSTM	0.97	0.1
k-fold validation	0.935	0.17

We ran the same model on the balanced datasets and performed k-fold validation:

Figure 2. BiLSTM and k-fold validation for different balanced versions of the dataset



The results show a steady increase in performance when more data are added and a respective decrease in the error loss rate. The k-fold results show comparable performance across the balanced datasets, which seems to show that the model is generalizing well.

6. Error analysis and limitations

We took a random sample of 20 miss-classified examples for each dialect and for each ML model. The results are quite interesting. In algorithms where performance is quite low, most of the miss-classified examples are genuine cases of miss-classification, where the algorithm has not managed to find enough distinctive features to distinguish the dialects. Consider the following example, a misclassified example from the SGD algorithm:

Text: Αυτοί πόρχουνδαν για καμνιά κούπα πιρνούσαν μbρουστά τ τουν gαλημέρζαν

Correct Dialect: northern

Predicted Dialect: cypriot

In the above, the classifier clearly cannot distinguish between northern varieties and Cypriot, even though the example contains typical features of northern varieties that do not appear in Cypriot (front vowel raising and front vowel loss). On the other hand, consider the following:

Text: ότι όλα έχουν τη θέση τους και την αξία

Correct Dialect: Cypriot

Predicted Dialect: Greek

This example represents a misclassification of a different kind. The mistake here is due to incomplete cleaning of the dataset. The example in question is clearly in Modern Greek that ended up in the Cypriot dataset. For example it could be part of a response to a blog post written in Cypriot (the post) in Standard Greek. In effect, the classifier is right here, pointing to incomplete cleaning in this case. In future work, we will use this type of error analysis based on the DI task to further clean the dataset and enhance its validity and reliability.

A further limitation of our work concerns the nature and type of data found in the dataset, in effect also connecting to the previous issue of data cleaning. We have created a dataset that is coarse-grained, in the sense that different varieties of the same dialect, different genres\registers\varieties in\of the dialect, have collapsed in one category. For example, in the case of Cypriot Greek, most of the data are collected from blog posts, which correspond to a specific text type, whereas in Cretan, a lot of the data come from translations of Ancient Greek tragedies and comedies into Cretan, and musical declamations (mantinades). However, despite this obvious limitation, we believe that the dataset can be used as a first basis to create more fine-grained datasets, where these considerations are taken into account.

7. Conclusion and future work

We have presented a comprehensive dataset of Greek dialectal data. Specifically, we collected freely available data from Pontic, Cretan, Northern varieties, and Cypriot Greek, performed light preprocessing, and compiled the first dataset of this magnitude for the computational study of Greek dialects. We conducted a simple Dialect Identification task using this dataset. The results indicate that the dialects possess distinctive characteristics that can be identified even by very basic ML algorithms, leading to good performance on the task. Notably, a simple BiLSTM model outperformed traditional ML algorithms, even with smaller data samples. The error analysis revealed that many errors stem from insufficient data cleaning. We are in the process of enhancing the cleaning process based on these findings and, also, further enrich the dataset with more data freely available on the web. Extending the dataset with data from more Modern Greek dialects is also something that we

are looking at. Given the recent development of the LLM Meltemi[7] (Roussis et al., 2025) that has been trained on Modern Greek data, we plan to make use of this resource to fine-tune Meltemi for dialectal text generation using our data. In general, we aspire to offer the community a dialectal dataset that can be used/extended/modified to be used for other dialectal NLP tasks.r.

Acknowledgements

The Special Account for Research Funding of the Technical University of Crete is thanked for funding part of this research (grant number: 11218). Stergios Chatzikyriakidis also gratefully acknowledges funding from the TALOS-AI4SSH ERA Chair in Artificial Intelligence for Humanities and Social Sciences grant (Grant agreement ID: 101087269). Stergios Chatzikyriakidis is also partially funded by the European Union (ERC ADG, PhylProGramm, 101096554). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them. We also want to thank the following people for their help with this project: a) Christodoulos Christodoulou for kindly giving us access to his corpus of Kozani Greek, b) Michalis Sfakianakis for kindly giving us access to his translations of numerous ancient Greek tragedies and comedies into Cretan Greek, and c) postgraduate students Irini Amanaki, Irini Giannikouri, Efrosini Skoulataki, Erofili Psaltaki, Valentini Mamatzaki, Vassiliki Katsouli and Hara Soupiona for their help in the initial stages of identifying dialectal data on the net.

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[7] https://huggingface.co/ilsp/Meltemi-7B-v1

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